



Time-dependent characterization and constitutive modeling of 3D-printed polymers

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Introduction

Tango Black is a polymer from the family of Tango.
It has various applications notably in testing.
Modeling the material is a must to run simulations of testing experiments.

	ASTM	Metric
Tensile Strength	D-412	1.8-2.4 MPa
Elongation at Break	D-412	45-55%
Compressive Set	D-395	0.5-1.5%
Shore Hardness (A)	D-2240	60-62 Scale A
Tensile Tear Resistance	D-624	3-5 Kg/cm
Polymerized density	D-792	1.14-1.15 g/cm ³



Content

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 - a. Viscoelasticity
 - b. Tensile Testing
- II. Characterization and constitutive modeling of Tango Black
 - a. Experimental setups and Results
 - b. Constitutive modeling and Simulation
 - c. Fracture Study
- III. Application: Compression of Tango Black Hexagonal Structures
 - a. Experimental setup
 - b. Results

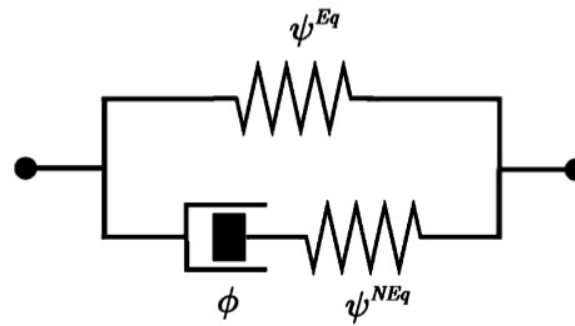
Theoretical Background

Viscoelasticity



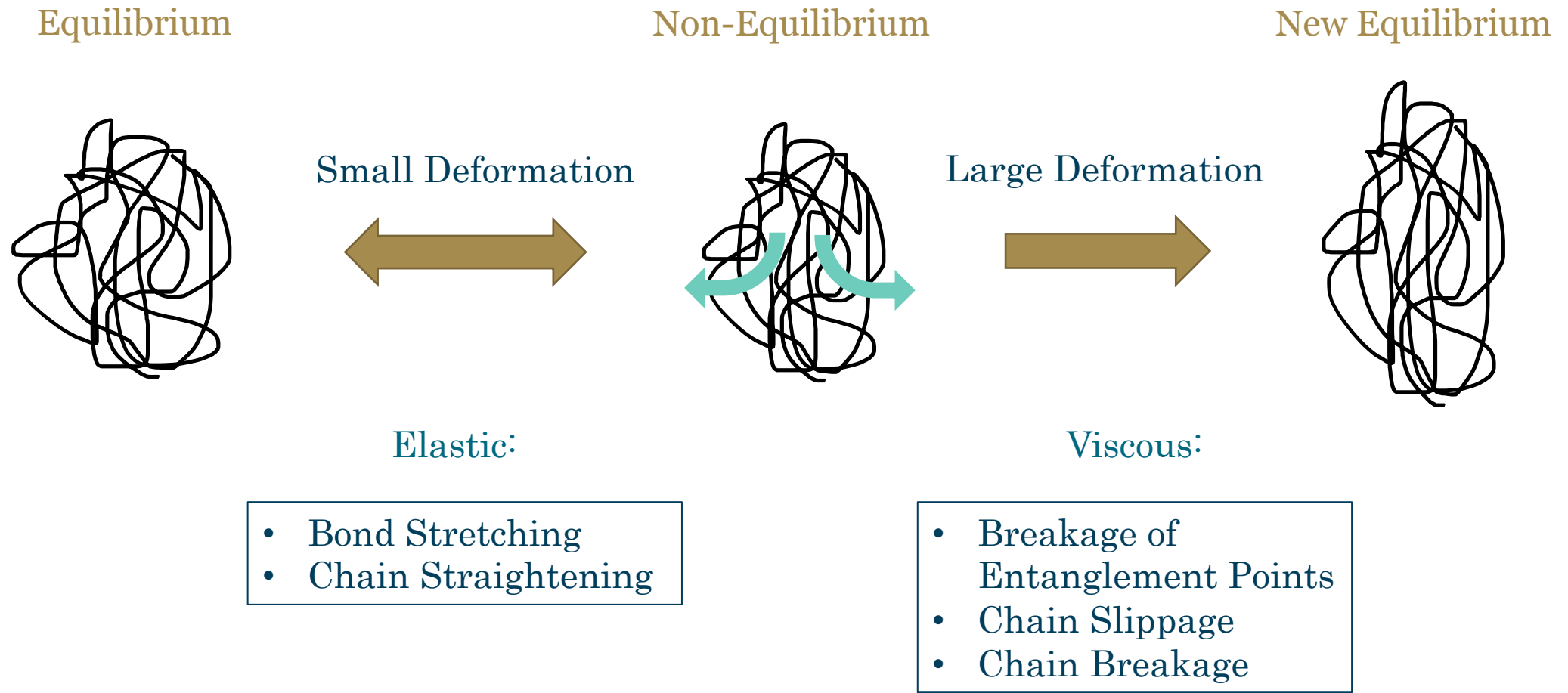
Viscoelastic material behavior is a combination of elastic and viscous behavior.

This is a rheologic model of rubber viscoelastic behavior, “Generalized standard model”.

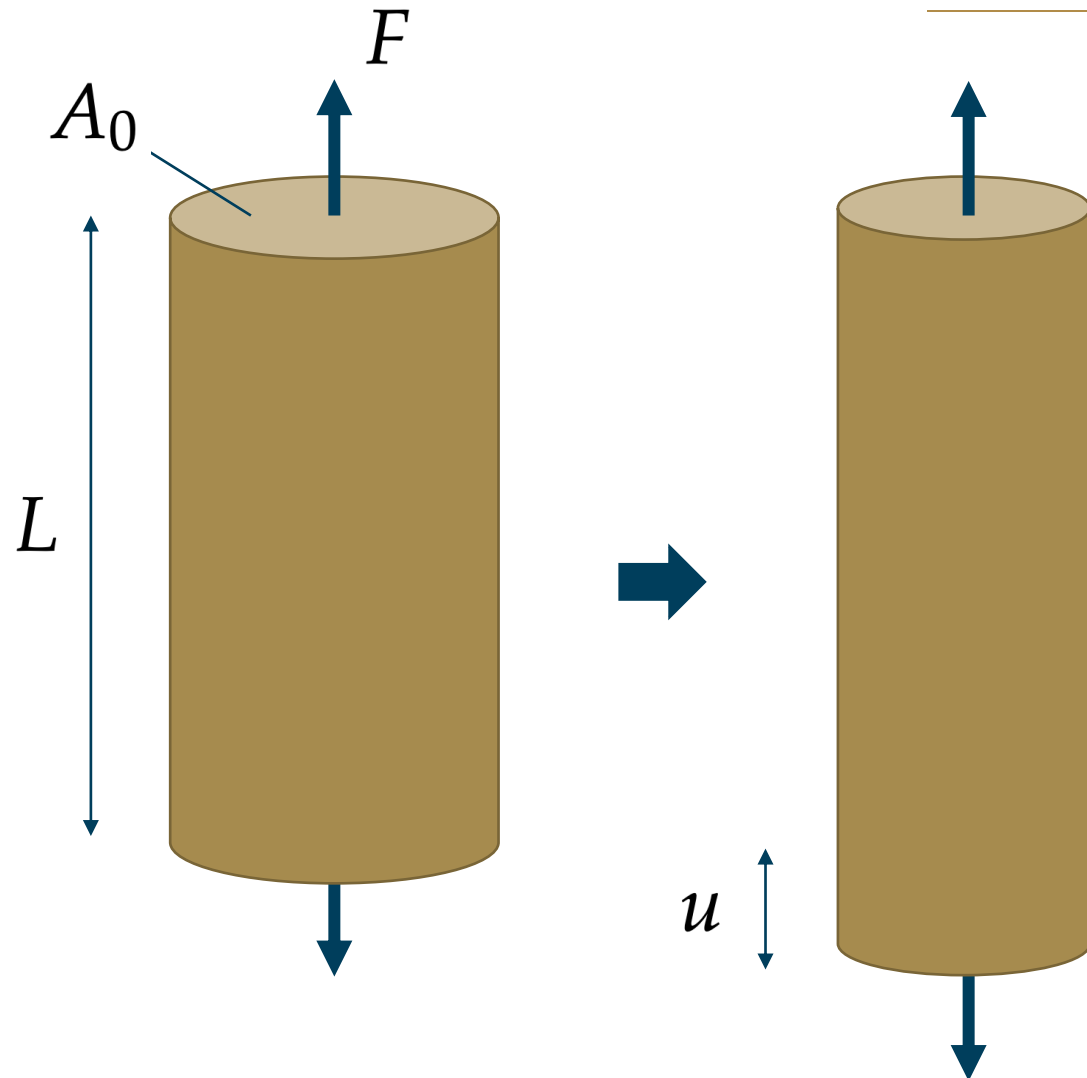


We will be interested in viscoelasticity at finite strain (large).

Viscoelasticity of Polymers



Tensile Testing



Engineering or Nominal Stress

$$S = \frac{F}{A_0}$$

Engineering or Nominal Strain

$$\lambda - 1 = \frac{u}{L}$$

Boundary conditions for uni-axial strain-controlled loading:

$$\mathbf{s} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & s_{33} \end{pmatrix} \quad \mathbf{F} = \begin{pmatrix} \frac{1}{\sqrt{\lambda}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda}} & 0 \\ 0 & 0 & \lambda \end{pmatrix} \quad \mathbf{F}^v = \begin{pmatrix} \frac{1}{\sqrt{\lambda_v}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda_v}} & 0 \\ 0 & 0 & \lambda_v \end{pmatrix}$$

Characterization and Constitutive Modeling of Tango Black

Sample Design

3D printing

3D printing or additive manufacturing is the process of building a three-dimensional structure from a CAD model.



3D Printer Objet Eden 260VS



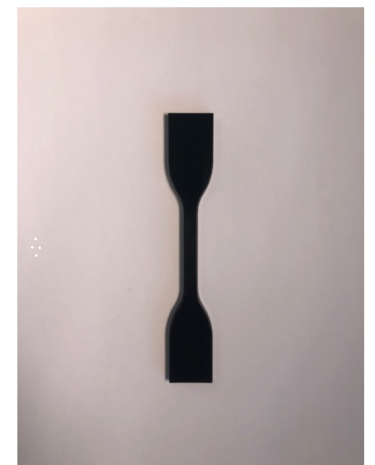
(A) Geometry ①



(B) Geometry ②



(C) Geometry ③



(D) Dogbone

Sample Design

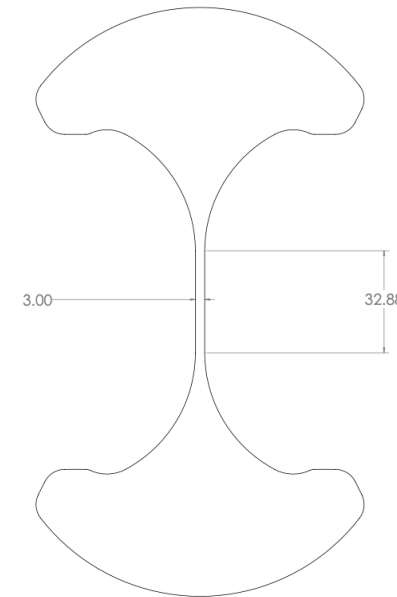
We started by using a classic dogbone specimen.



Drawbacks

- Breaks at the non-linear part.
- Slips out of the clamps.

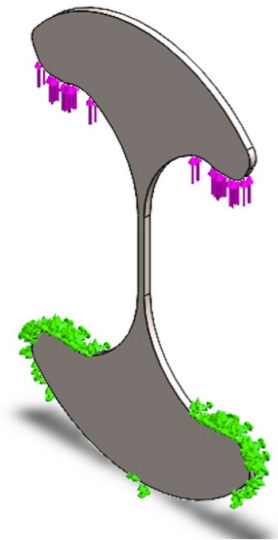
Then, we changed the geometry to a band specimen.



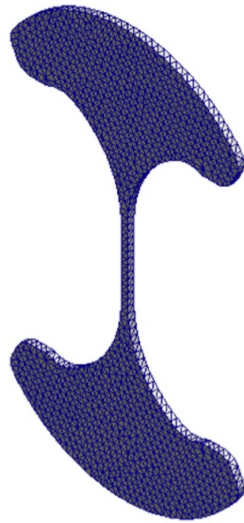
Useful Length: 78 mm
Cross section area: 18 mm²

Sample Design

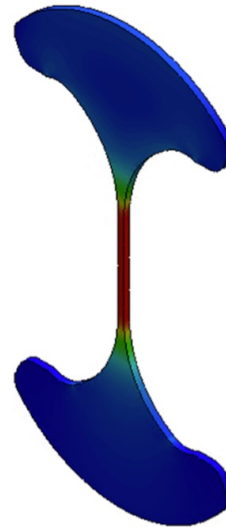
SolidWorks Simulation



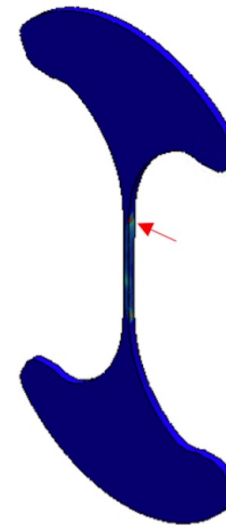
(A) Boundary conditions and force initialization



(B) Meshing

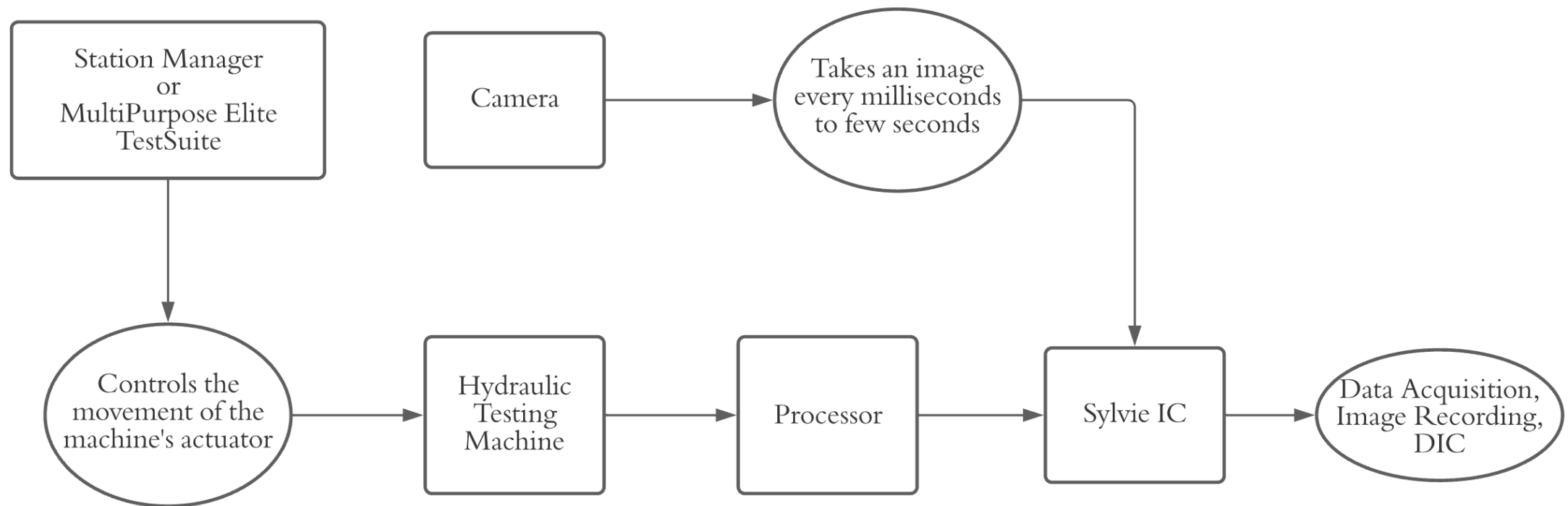


(C) Von Mises Stress plot

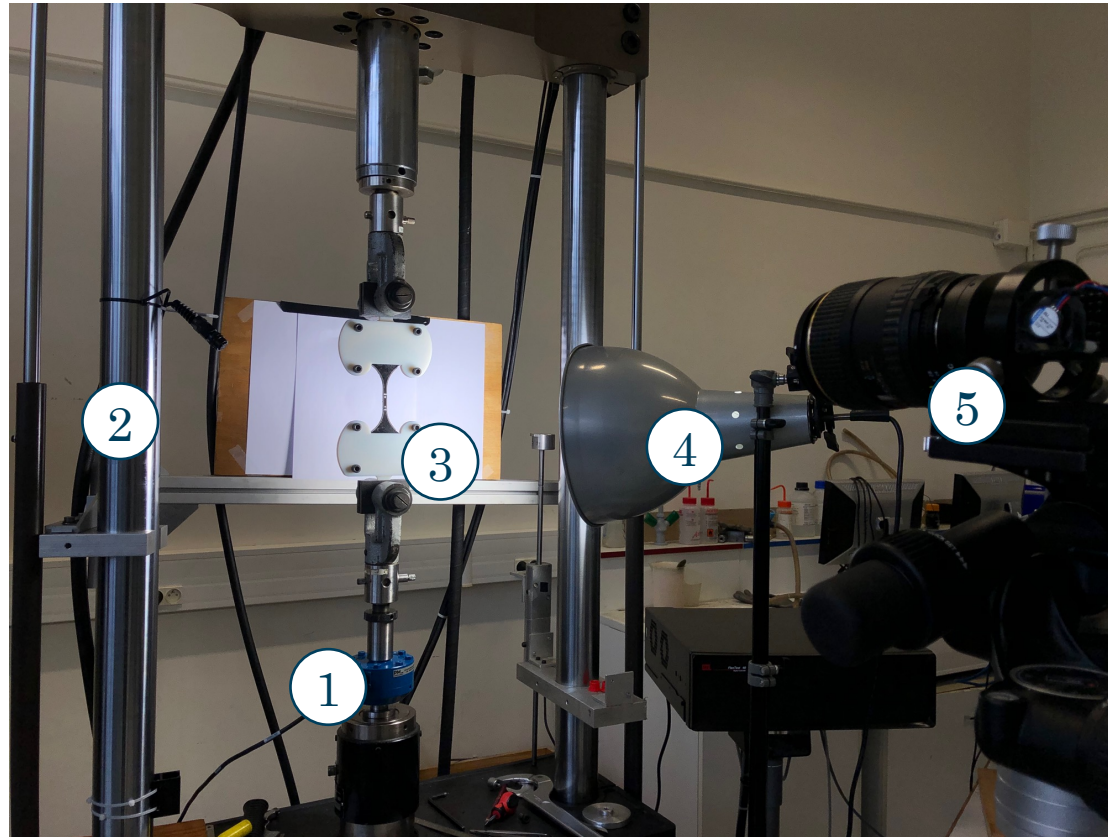


(D) Location of maximal stress

Diagram of Experimental Setup



Tensile and Cyclic Testing: Experimental Setup



1. Load Cell 1500 N 2. MTS 250kN 3. Grips 4. Lamp
5. Camera

Using the Hydraulic Testing Machine MTS 250kN and the Geometry 1:

Maximal Strain rate is:

$$\frac{30\text{mm/s}}{78\text{mm}} \approx 0.4\text{s}^{-1}$$

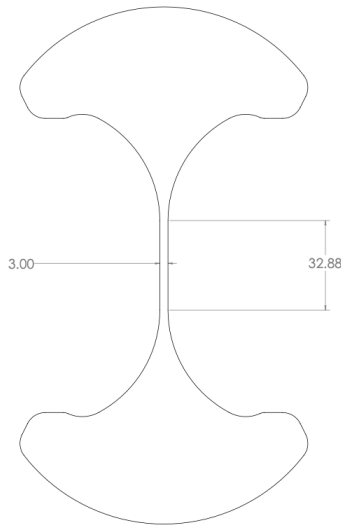
Characteristics	MTS 100kN	MTS 250kN
Load range	± 100 kN	± 250 kN
Stroke	± 100 mm	± 80 mm
Actuator maximum speed	100 mm/s	30 mm/s

Using the Hydraulic Testing Machine MTS 100kN and the Geometry 1:

Maximal Strain rate is:

$$\frac{100\text{mm/s}}{78\text{mm}} \approx 1.3\text{s}^{-1}$$

Tensile and Cyclic Testing: Increasing the Strain Rate

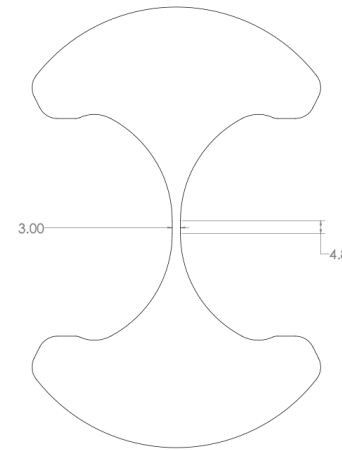


Using the Hydraulic Testing Machine MTS 100kN and the Geometry 1:

Maximal Strain rate is:

$$\frac{100\text{mm/s}}{78\text{mm}} \approx 1.3\text{s}^{-1}$$

Useful Length: 78 mm
Cross section area: 18 mm²



Using the Hydraulic Testing Machine MTS 100kN and the Geometry 2:

Maximal Strain rate is:

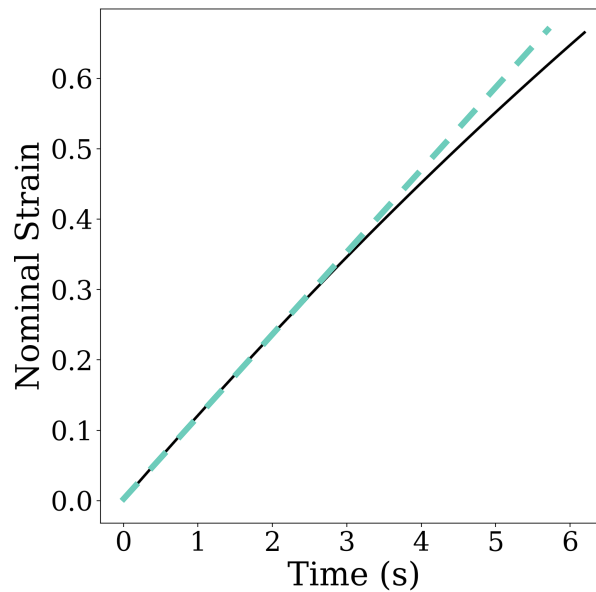
$$\frac{100\text{mm/s}}{45\text{mm}} \approx 2.2\text{s}^{-1}$$

Useful Length: 45 mm
Cross section area: 18 mm²

Tensile and Cyclic Testing: Limitations

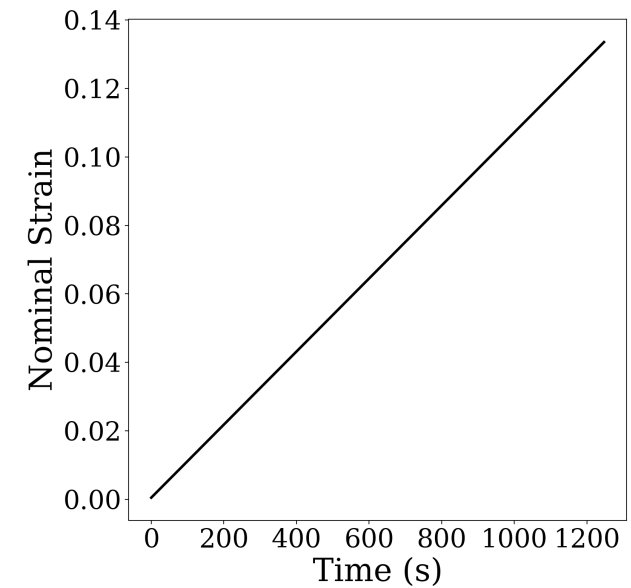
Non-linear strain for high strains at high strain rates

Strain Rate: 0.1 s^{-1}



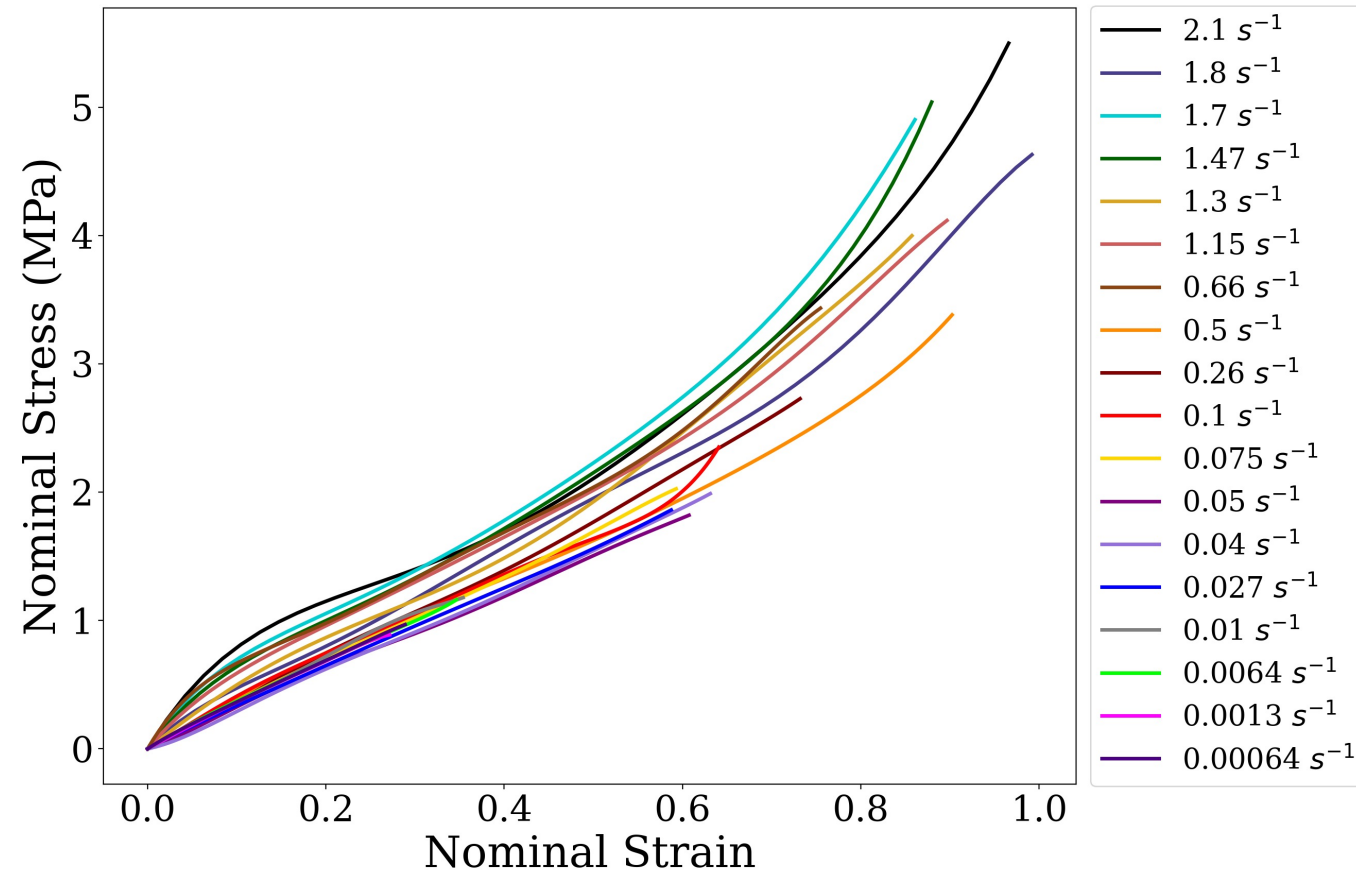
Speed: $\times 1$

Strain Rate: 0.0001 s^{-1}



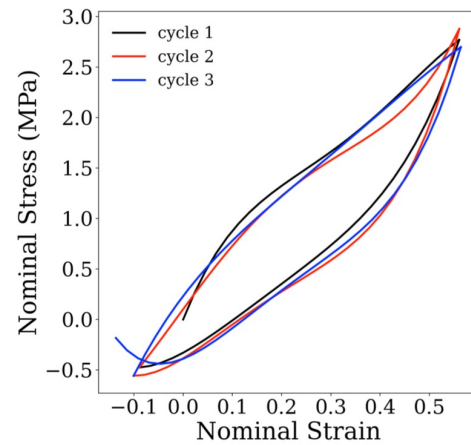
Speed: $\times 120$

Tensile and Cyclic Testing: Results

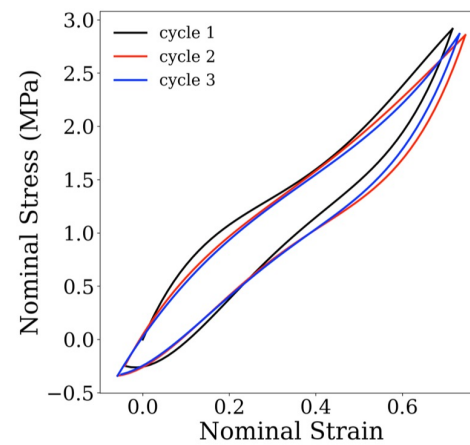


Strain Rate	Young's Modulus	Poisson Ratio
10^{-1}	4.1	0.46
10^{-2}	4.007	0.4766
10^{-3}	3.914	0.4879
10^{-4}	3.818	0.4956

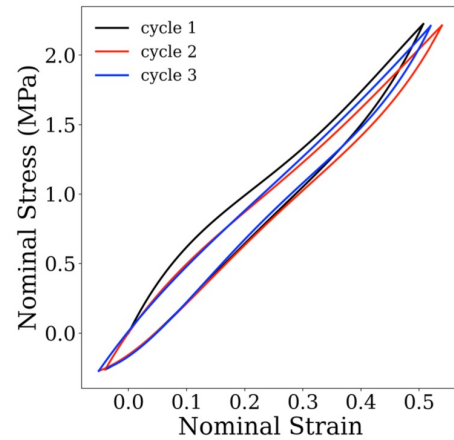
Tensile and Cyclic Testing: Results



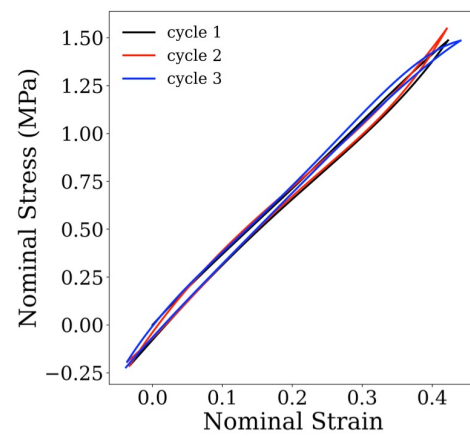
(A) Strain rate: 2.2 s^{-1}



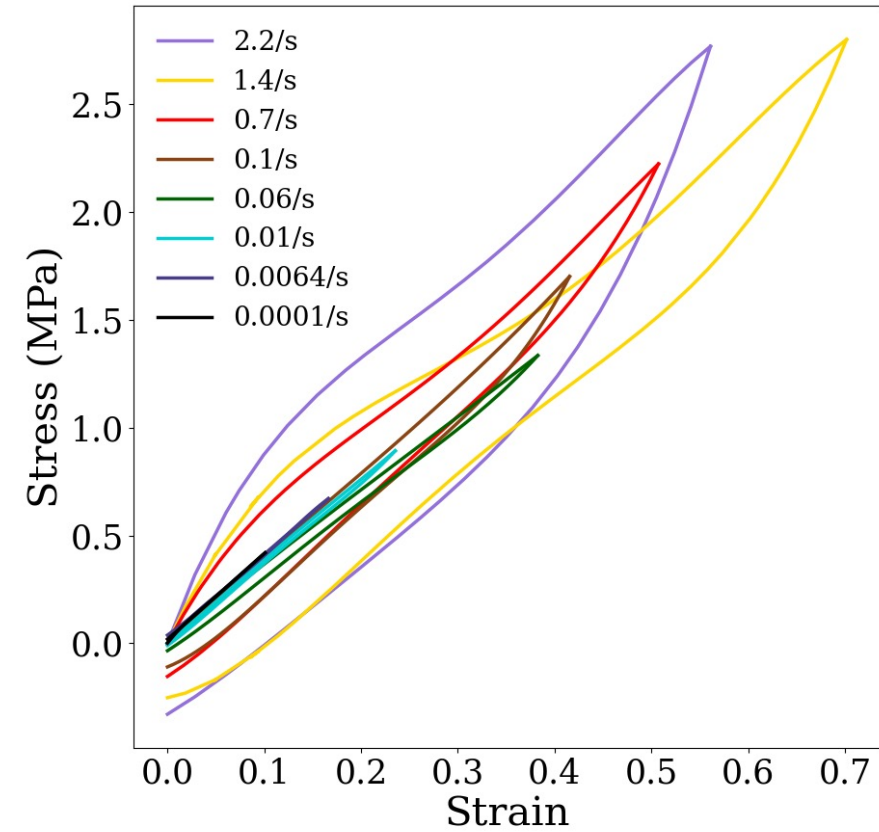
(B) Strain rate: 1.4 s^{-1}



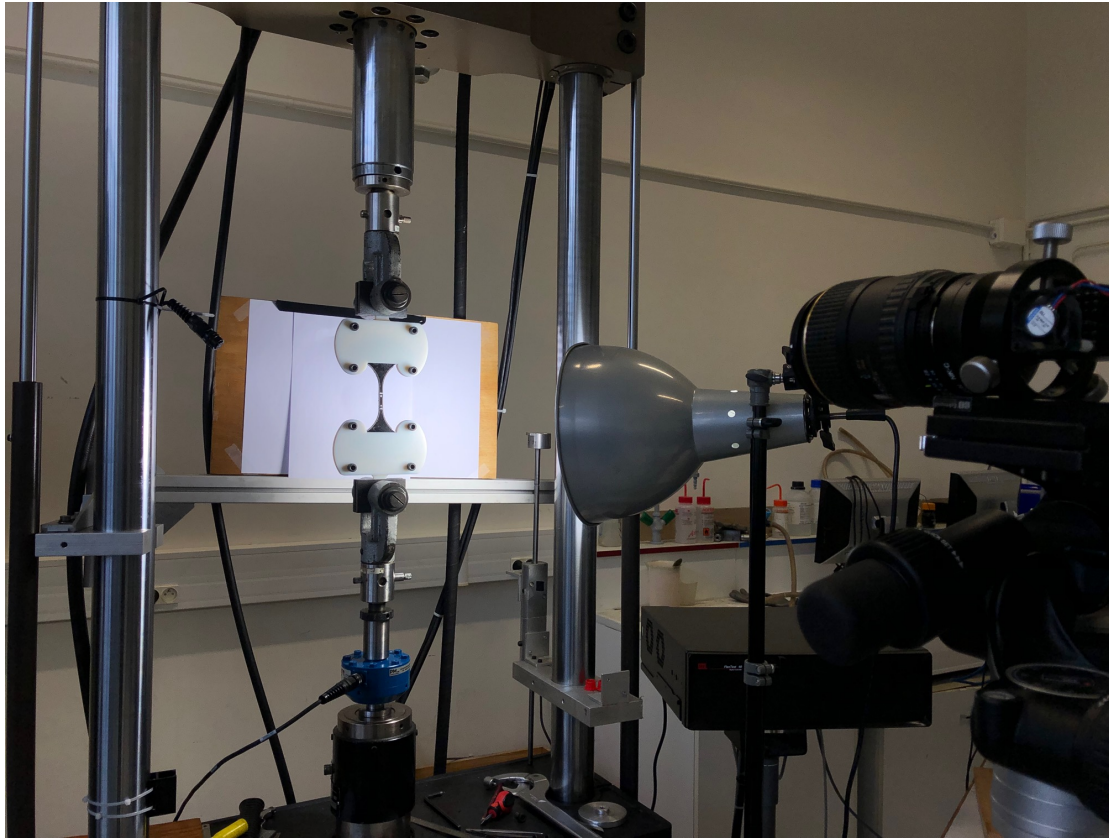
(C) Strain rate: 0.7 s^{-1}



(D) Strain rate: 0.06 s^{-1}



Relaxation Testing: Experimental Setup



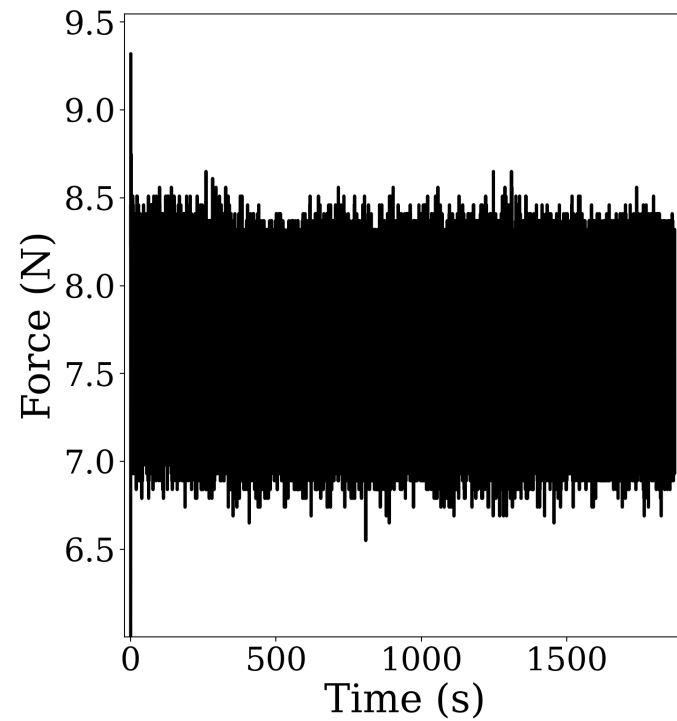
Relaxation tests were realized at a constant deformation level.



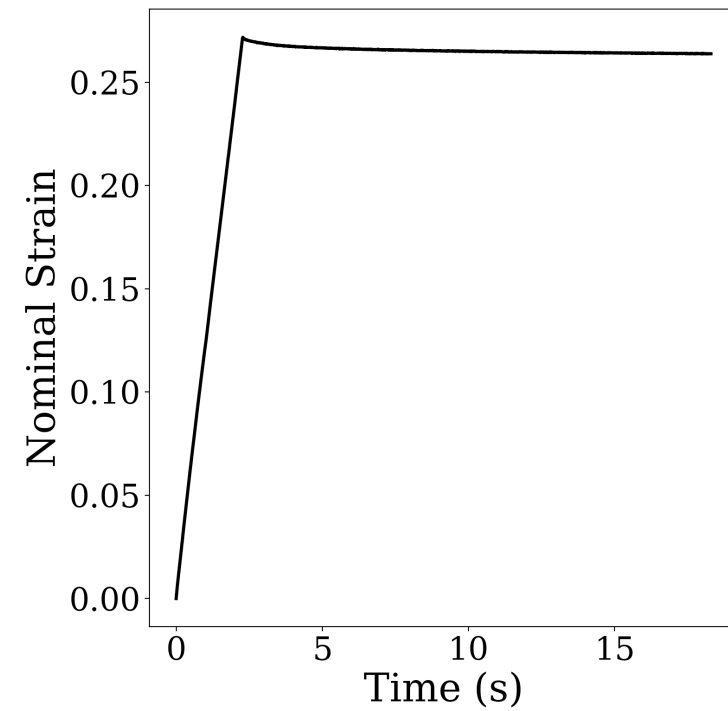
Tango Black is semi-brittle.
Relaxation tests should be realized at a relatively low force to avoid fracture.

Relaxation Testing: Challenges

1. Noisy measurements



2. “Relaxation” in Strain

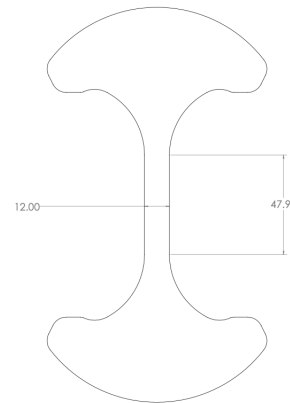


Relaxation Testing: Challenges

1. Noisy measurements: Solution



Increase the cross sectional area to increase the force measured

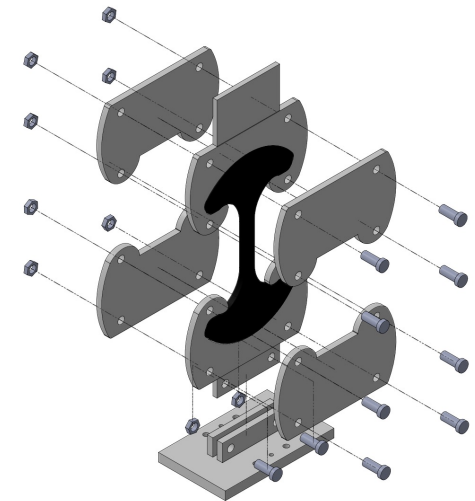
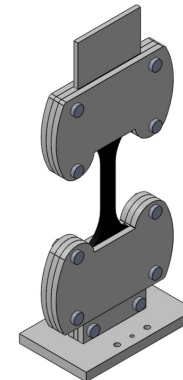


Useful Length: 120 mm

Cross section area: $6\text{mm} \times 12\text{mm} = 72\text{ mm}^2$

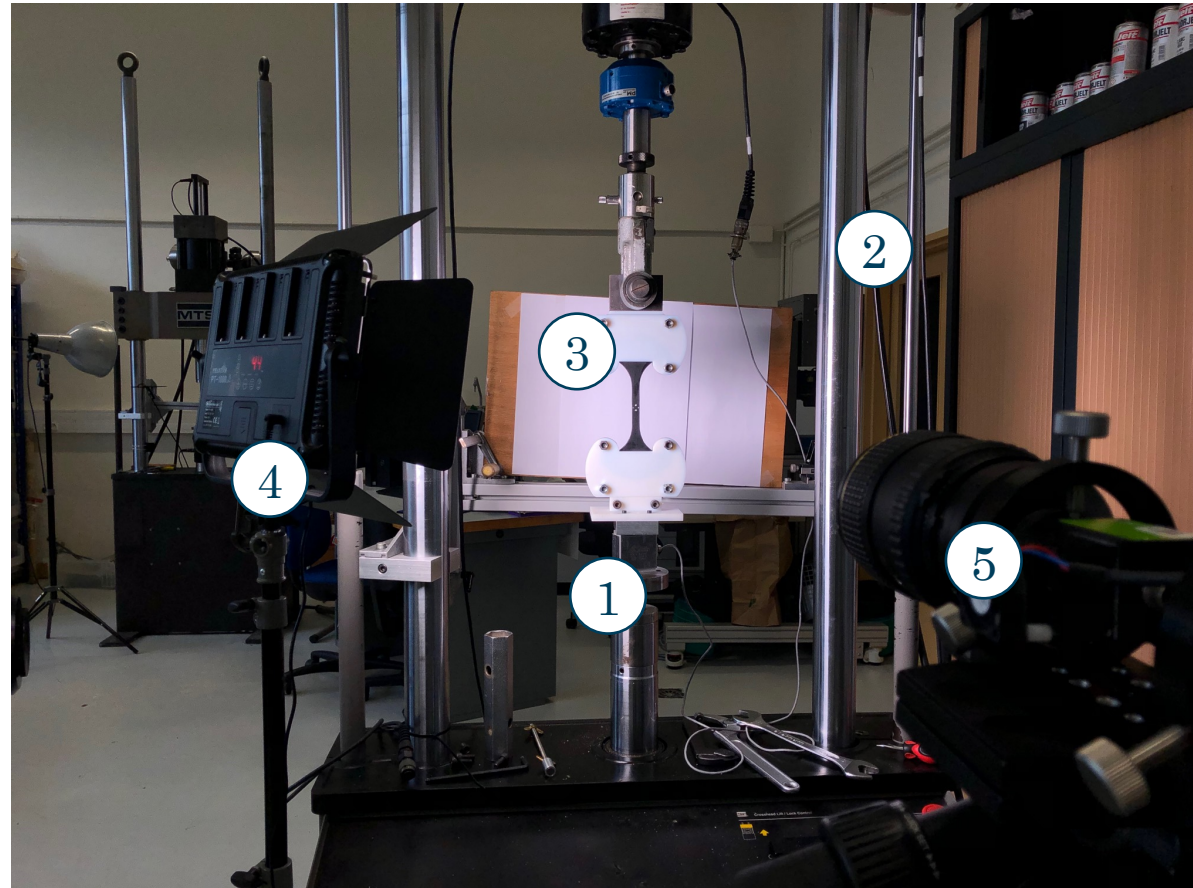


Change the 1500 N Load Cell to a 200 N Load Cell (precision increased, requires a change of setup)



Relaxation Testing: Challenges

1. Noisy measurements: Solution



1. Load Cell 200 N 2. MTS 250kN 3. Grips 4. Lamp
5. Camera

Relaxation Testing: Challenges

2. “Relaxation” in Strain: Solution

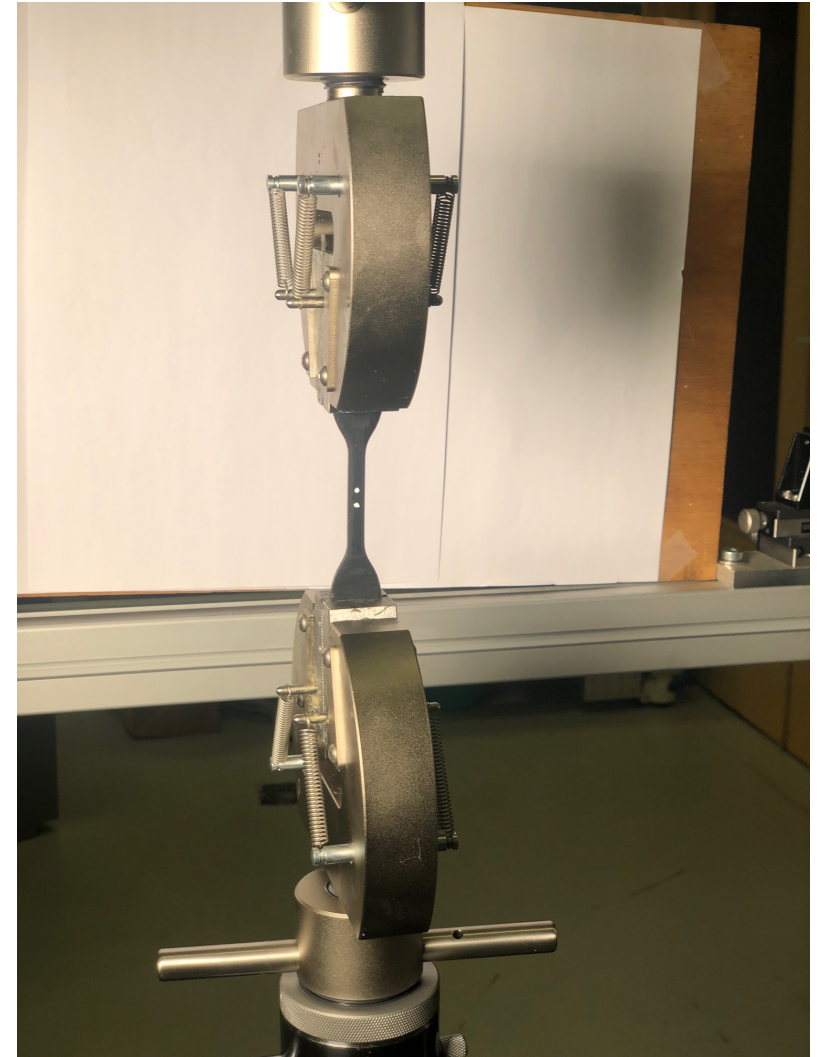


Use a dogbone specimen instead of a band geometry



Useful Length: 55 mm

Cross section area: $6\text{mm} \times 3.3\text{mm} = 19.8\text{ mm}^2$



Relaxation Testing: Challenges

2. “Relaxation” in Strain: Solution

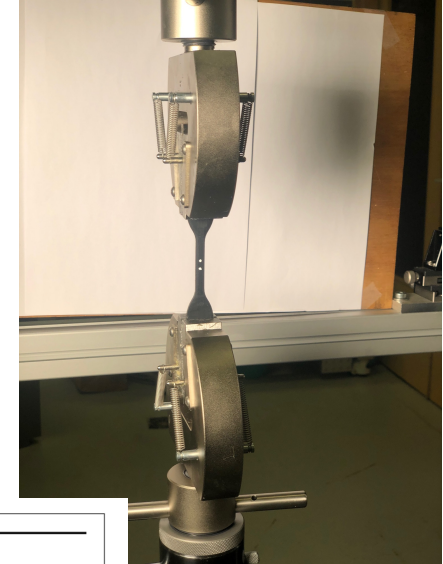
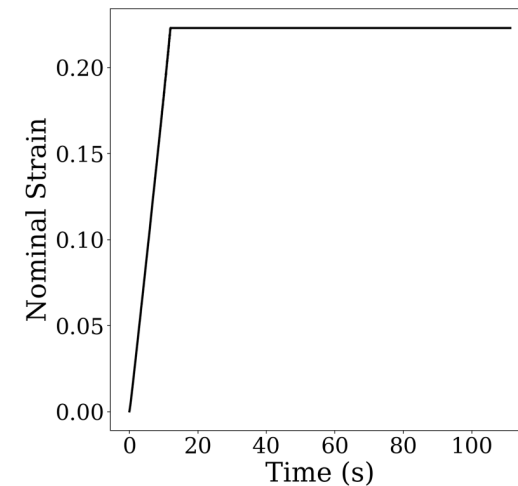


Use a dogbone specimen instead of a band geometry

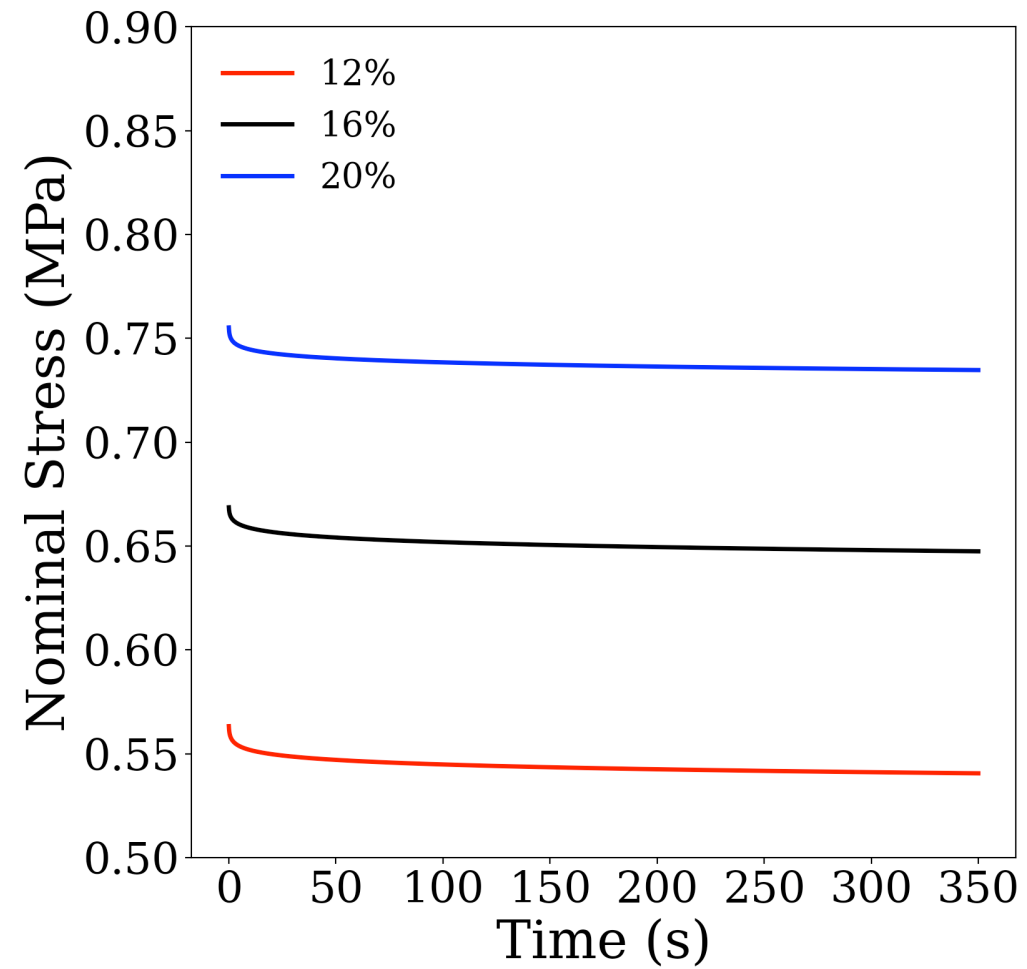


Useful Length: 55 mm

Cross section area: $6\text{mm} \times 3.3\text{mm} = 19.8\text{ mm}^2$



Relaxation Testing: Results



The two-potential constitutive modeling of rubber viscoelastic materials proposed by Lopez-Pamies

$$\psi^{\text{Eq}}(\mathbf{F}) = \begin{cases} \frac{3^{1-\alpha_1}}{2\alpha_1} \mu_1 [I_1^{\alpha_1} - 3^{\alpha_1}] + \frac{3^{1-\alpha_2}}{2\alpha_2} \mu_2 [I_1^{\alpha_2} - 3^{\alpha_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases}$$

$$\psi^{\text{NEq}}(\mathbf{F}^e) = \begin{cases} \frac{3^{1-a_1}}{2a_1} m_1 [I_1^{a_1} - 3^{a_1}] + \frac{3^{1-a_2}}{2a_2} m_2 [I_1^{a_2} - 3^{a_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases}$$

$$\mathbf{S} = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r I_1^{\alpha_r-1} \right] \mathbf{F} + \left[\sum_{r=1}^2 3^{1-a_r} m_r (\mathbf{C} \cdot \mathbf{C}^{\mathbf{v}-1})^{a_r-1} \right] \mathbf{F} \mathbf{C}^{\mathbf{v}-1} - p \mathbf{F}^{-T}$$

$$\dot{\mathbf{C}}^{\mathbf{v}} = \frac{\sum_{r=1}^2 3^{1-a_r} m_r (\mathbf{C} \cdot \mathbf{C}^{\mathbf{v}-1})^{a_r-1}}{3\eta_k} \left(\mathbf{C} - \frac{1}{3} (\mathbf{C} \cdot \mathbf{C}^{\mathbf{v}-1}) \mathbf{C}^{\mathbf{v}} \right)$$

$$S = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r \left(\frac{2}{\lambda} + \lambda^2 \right)^{\alpha_r-1} \right] \left(\lambda - \frac{1}{\lambda^2} \right) + \left[\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1} \right] \left(\frac{\lambda}{\lambda_v^2} - \frac{\lambda_v}{\lambda^2} \right)$$

Stress-deformation response for the uniaxial strain-controlled deformation

$$\dot{\lambda}_v = \frac{\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1}}{3\eta_k} \left(\frac{\lambda^2}{\lambda_v} - \frac{\lambda_v^2}{\lambda} \right)$$

λ_v is a solution of this ODE to be solved with Runge-Kutta method

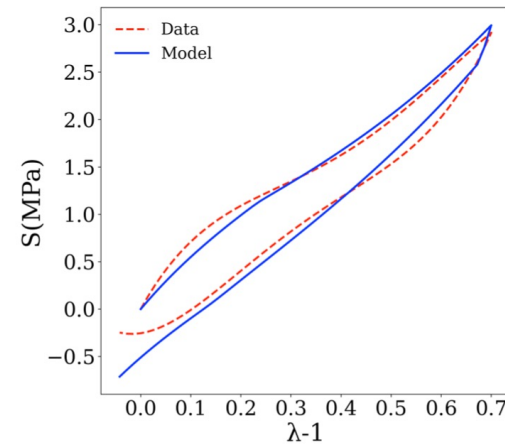
$$\eta_k = \eta_\infty + \frac{\eta_0 - \eta_\infty + k_1 \left[\left(\frac{2}{\lambda_v} + \lambda_v^2 \right)^{\beta_1} - 3^{\beta_1} \right]}{1 + (k_2 J_2^{\text{NEq}})^{\beta_2}}$$

$$J_2^{\text{NEq}} = \frac{1}{3} \left(\frac{\lambda_v}{\lambda} - \frac{\lambda^2}{\lambda_v^2} \right)^2 \left(\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1} \right)^2$$

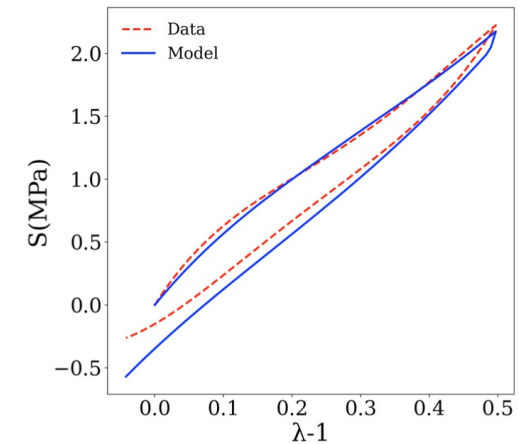
Model for isotropic incompressible elastomers proposed by Lopez-Pamies

Parameter	Fitting
μ_1	625.8 kPa
μ_2	625.8 kPa
α_1	2.3664
α_2	2.3668
m_1	362.8 kPa
m_2	362.7 kPa
a_1	-1.3595
a_2	-1.36
K_1	274 kPa s
η_0	4.7 kPa s
η_∞	0.3 kPa s
β_1	26.7236
β_2	0
K_2	18.9752 MPa ²

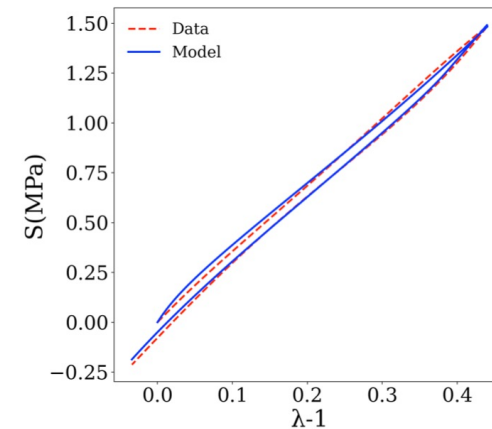
Sumsq = 3.56



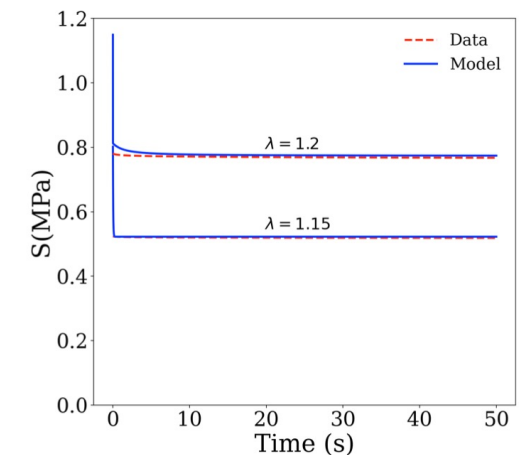
(A) Strain rate: 1.4 s^{-1}



(B) Strain rate: 0.7 s^{-1}

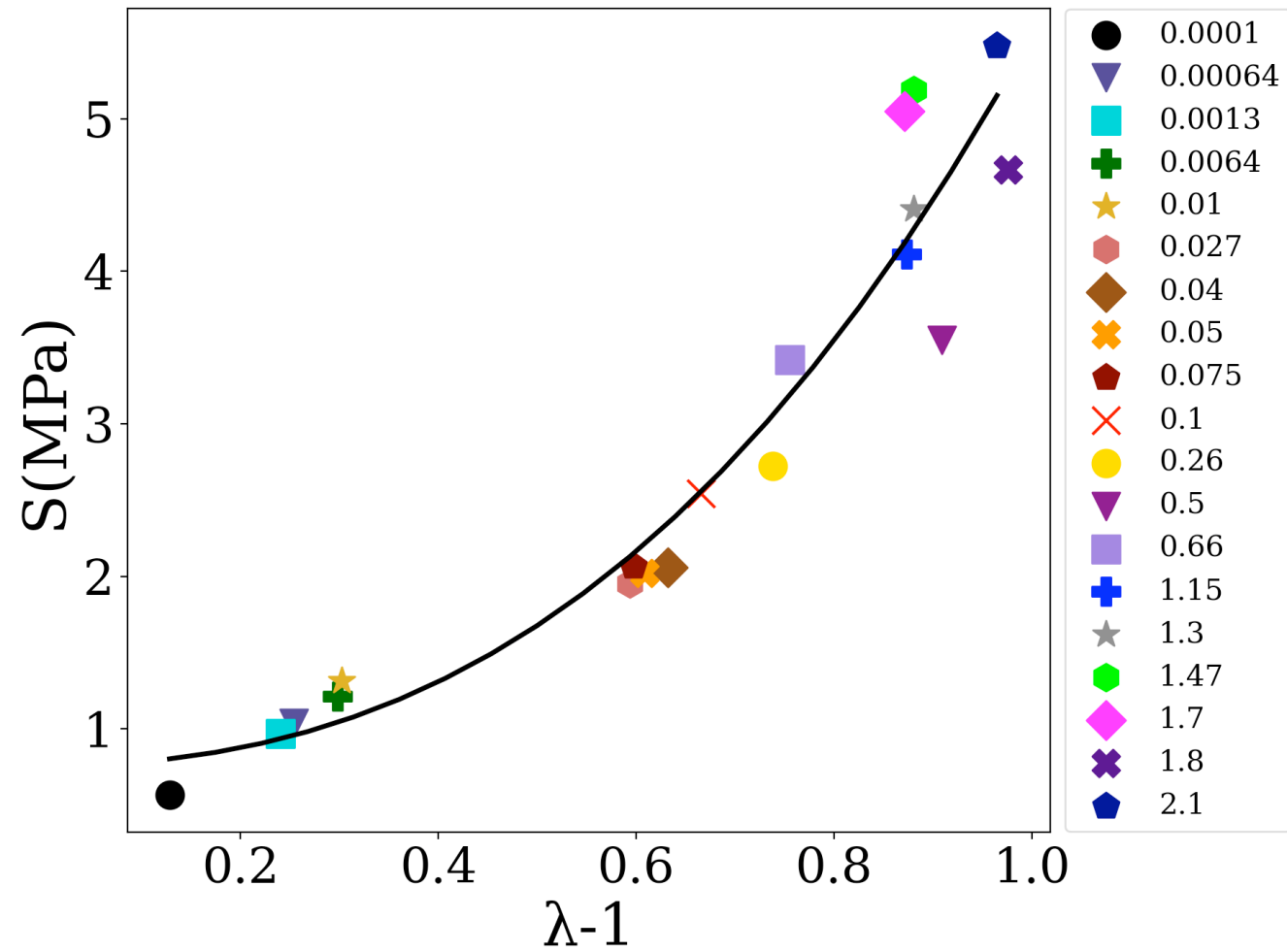


(C) Strain rate: 0.06 s^{-1}



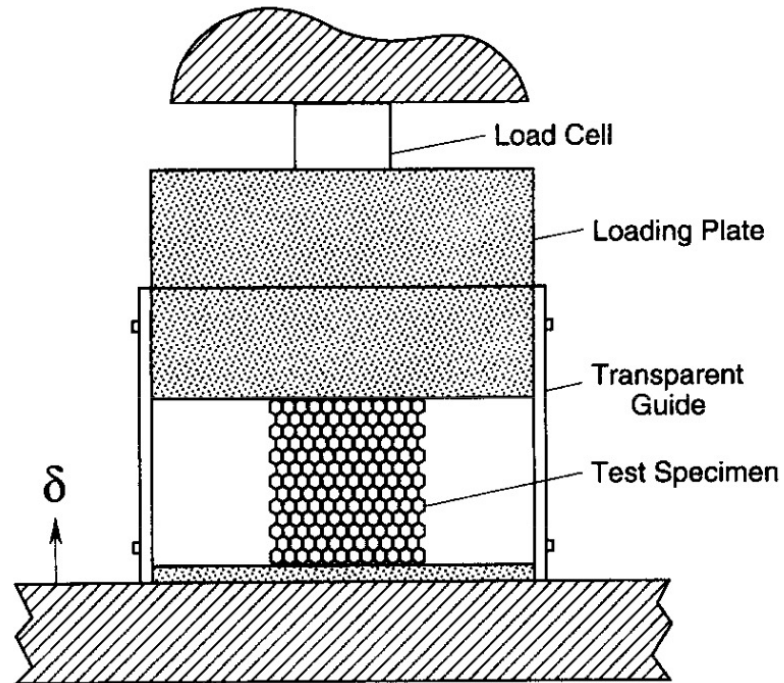
(D) Relaxation at 15% and 20%

Fracture Study



Application: Compression of Tango Black Hexagonal Structures

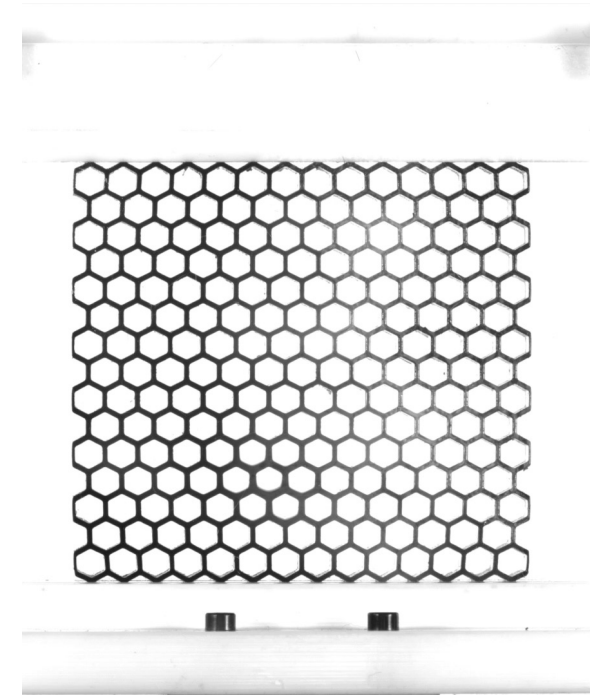
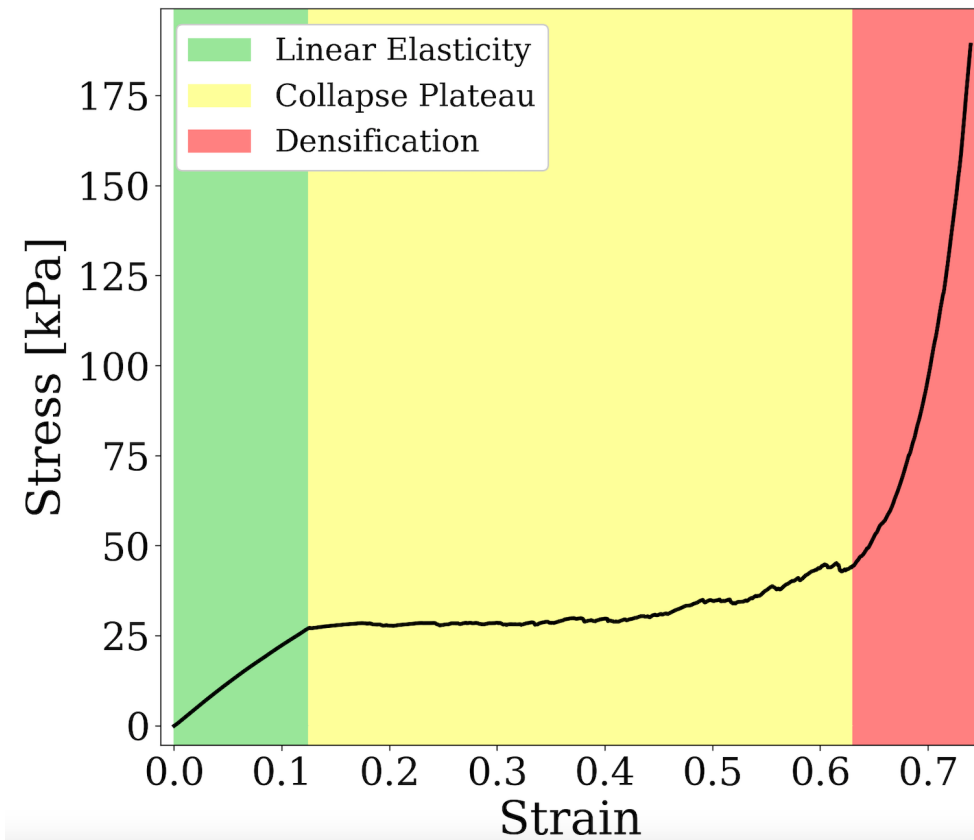
Experimental Setup



Parameters:

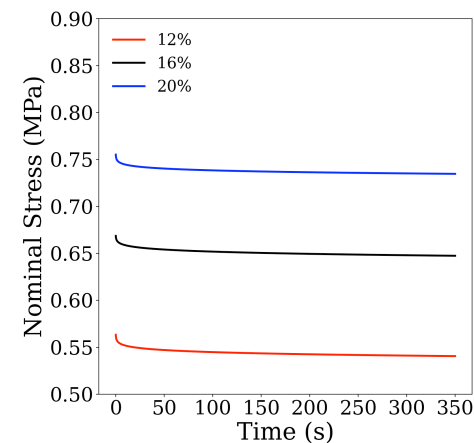
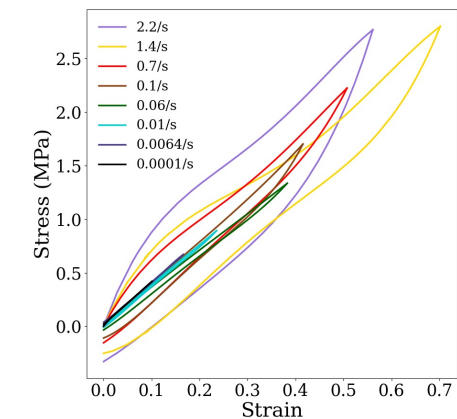
- Displacement control at displacement rate 0.01 mm/s
- Stopping criterion: force control
 - Porosity of 70%: stopped at 190 N
 - Porosity of 80%: stopped at 120 N
 - Porosity of 90%: stopped at 10 N

Results



Conclusion

Mechanical Testing

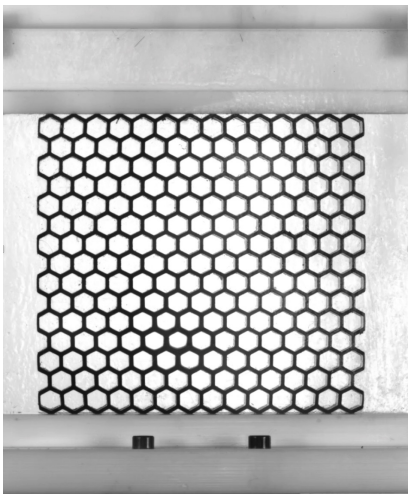


Constitutive modeling

$$\psi^{\text{Eq}}(\mathbf{F}) = \begin{cases} \frac{3^{1-\alpha_1}}{2a_1} \mu_1 [I_1^{\alpha_1} - 3^{\alpha_1}] + \frac{3^{1-\alpha_2}}{2a_2} \mu_2 [I_1^{\alpha_2} - 3^{\alpha_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases}$$
$$\psi^{\text{NEq}}(\mathbf{F}^e) = \begin{cases} \frac{3^{1-a_1}}{2a_1} m_1 [I_1^{a_1} - 3^{a_1}] + \frac{3^{1-a_2}}{2a_2} m_2 [I_1^{a_2} - 3^{a_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases}$$

Parameter	Fitting
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η_0	4.7 kPa s
η_∞	0.3 kPa s
β_1	26.7236
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K_2	18.9752 MPa ²

Application in structure testing



References

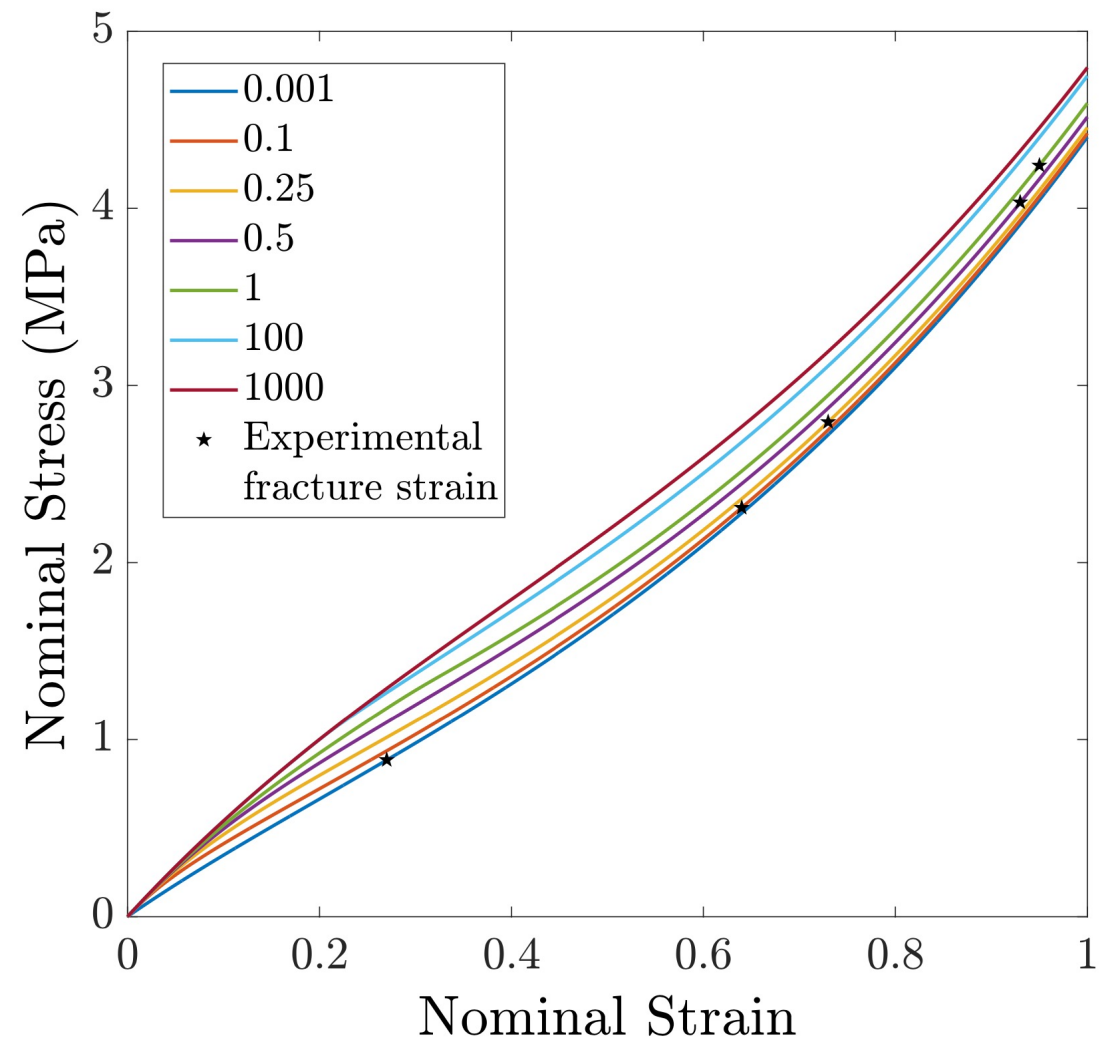
1. A. Kumar, O. Lopez-Pamies, On the two-potential constitutive modeling of rubber viscoelastic materials (2015) 1631-0721.
2. L. Gibson, M. Ashby, Cellular solids, Structure and Properties, 2nd Edition (1997).
3. S. D. Papka and S. Kyriakides, In-plane compressive response and crushing of honeycombs (1994) 0022-5096.
4. Stratasys, Tango, Polyjet Simulated Rubber Material.[Online] Available: https://www.stratasys.com/siteassets/materials/materials-catalog/polyjet-materials/tango/mss_pj_tango_0318a.pdf?v=48e1c7 [Accessed: 03/04/2022]
5. CADVision,ObjetEden26oVSparStratasys.[Online] Available: <https://www.cadvision.fr/imprimante-3d/imprimante-3d-objet-eden-260vs> [Accessed: 03/04/2022]
6. Stratasys, LWT Systemas, Impressora 3D Stratasys Objet Eden26oVS, Polyjet 1. [Online] Available: <https://www.lwtsistemas.com.br/produtos/impressora-3d-stratasys-objet-eden260vs-polyjet/> [Accessed: 03/04/2022]



Merci



Simulation



Neo-Hookean incompressible model

Equilibrium and non-equilibrium free energy functions

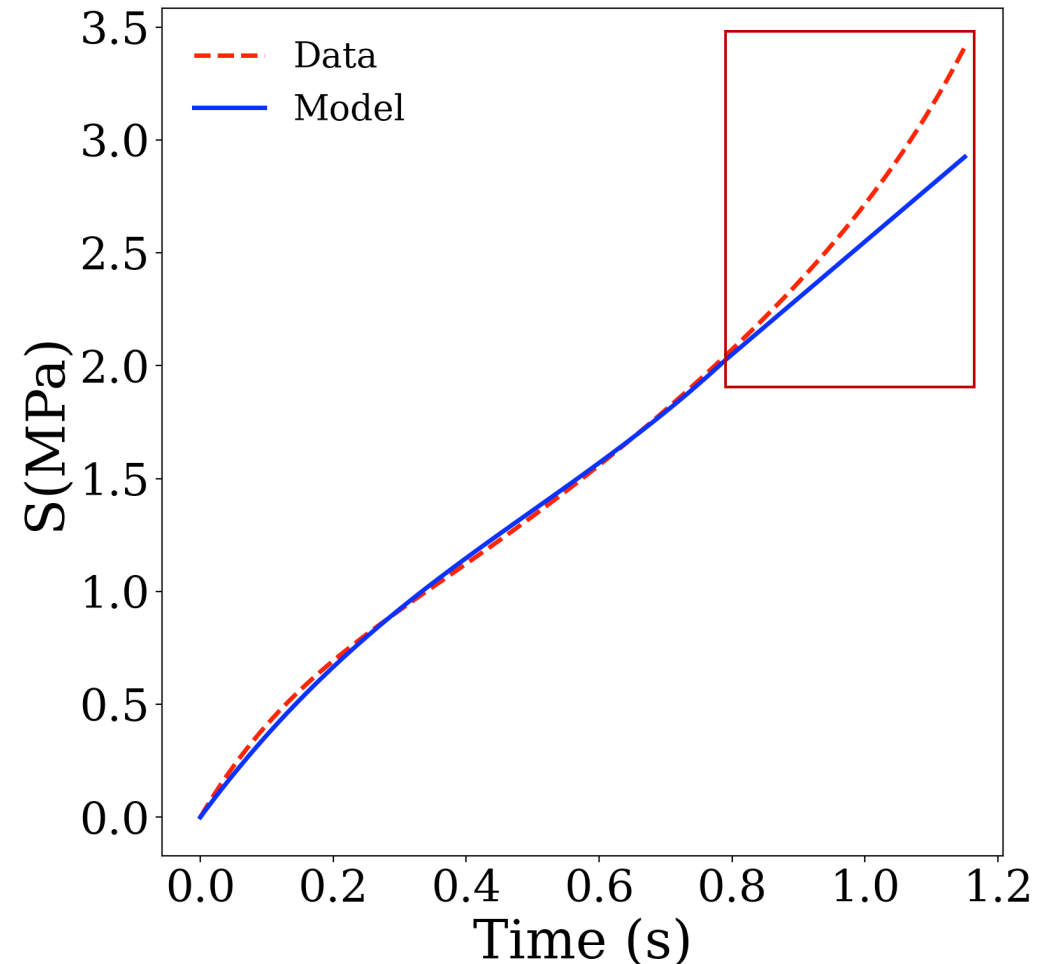
$$\psi^{\text{Eq}}(\mathbf{F}) = \frac{\mu_1}{2} (\mathbf{F} \cdot \mathbf{F} - 3) \text{ and } \psi^{\text{NEq}}(\mathbf{F}^e) = \frac{\mu_2}{2} (\mathbf{F}^e \cdot \mathbf{F}^e - 3)$$

Stress-deformation response for the uniaxial strain-controlled deformation:

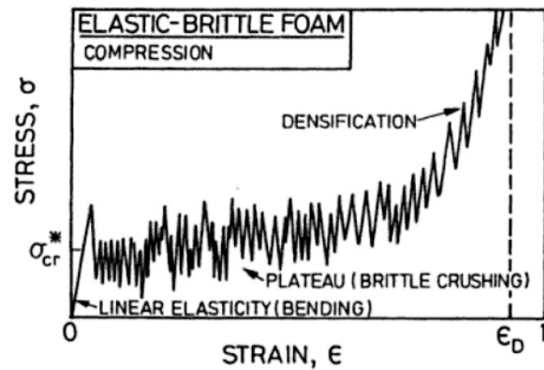
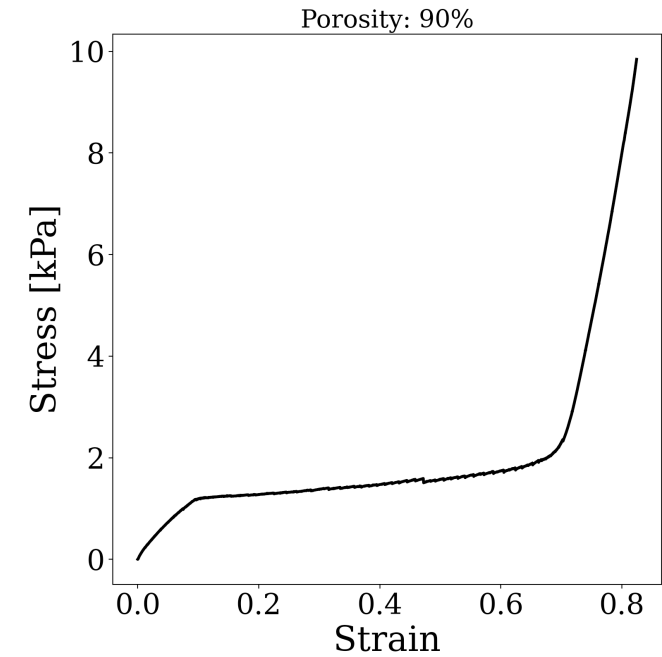
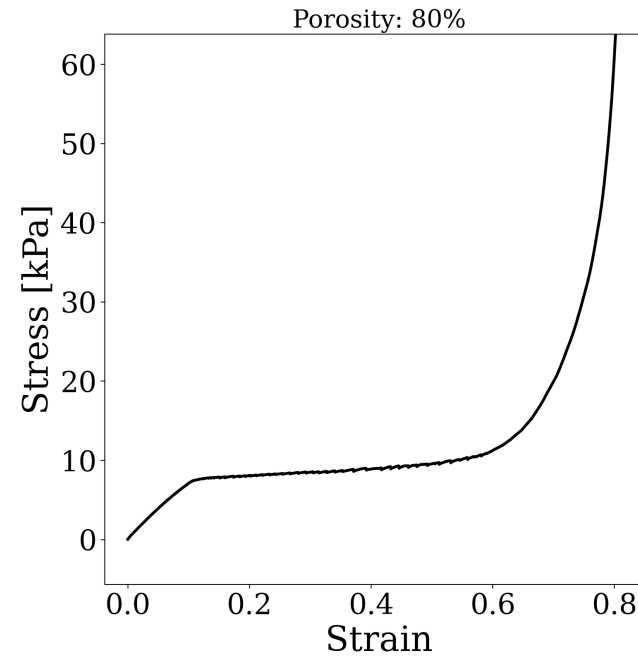
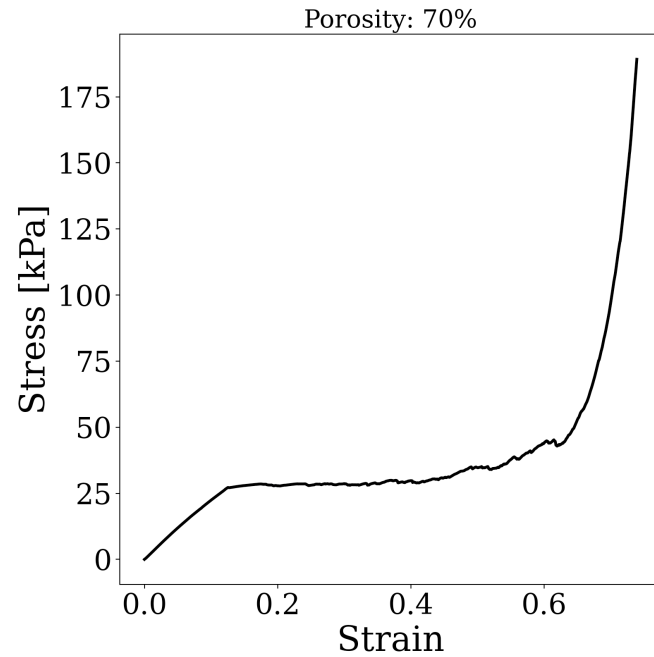
$$S = \mu_1 \left(\lambda - \frac{1}{\lambda^2} \right) + \mu_2 \left(\frac{\lambda}{\lambda_v^2} - \frac{\lambda_v}{\lambda^2} \right)$$

λ_v is a solution of this ODE to be solved with Runge-Kutta method:

$$\dot{\lambda}_v = \frac{\mu_2}{3\eta_k} \left(\frac{\lambda^3 - \lambda_v^3}{\lambda \lambda_v} \right)$$

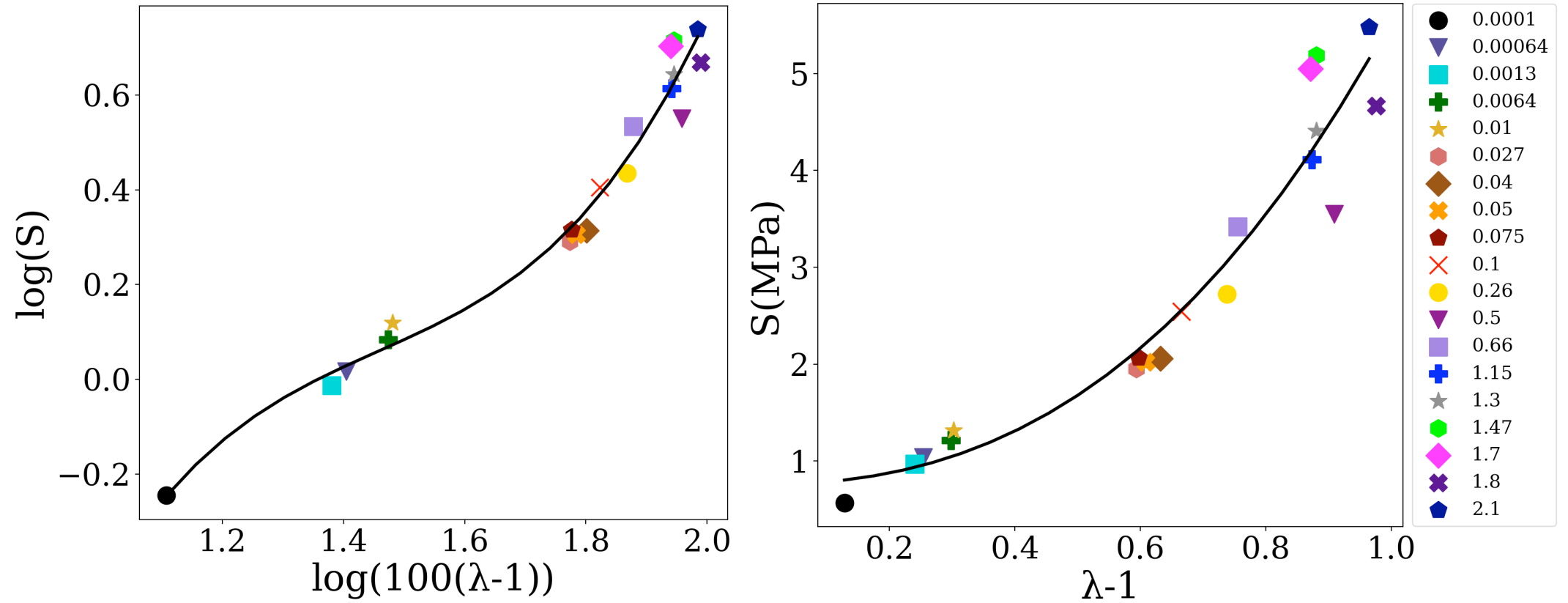


More Results for Part 3



ρ	E (kPa)	S_P (kPa)	ϵ_D
0.7	233.93 ± 5.66	30 ± 1.5	0.52 ± 0.05
0.8	97.21 ± 4.68	9 ± 1	0.6 ± 0.05
0.9	17.56 ± 2.37	1.6 ± 0.4	0.7 ± 0.05

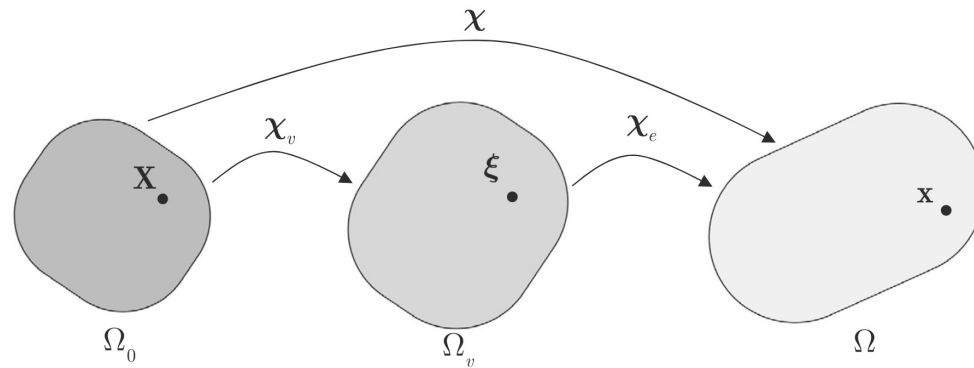
Fracture Study



Boundary Conditions

$$\mathbf{S} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & S_{33} \end{pmatrix} \quad \mathbf{F} = \begin{pmatrix} \frac{1}{\sqrt{\lambda}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda}} & 0 \\ 0 & 0 & \lambda \end{pmatrix} \quad \mathbf{F}^v = \begin{pmatrix} \frac{1}{\sqrt{\lambda_v}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda_v}} & 0 \\ 0 & 0 & \lambda_v \end{pmatrix}$$

Assumption



$$\mathbf{F} = \frac{\partial \chi}{\partial \mathbf{X}} = \frac{\partial \chi_e}{\partial \xi} \frac{\partial \chi_v}{\partial \mathbf{X}} = \mathbf{F}^e \mathbf{F}^v$$

The two-potential constitutive modeling of rubber viscoelastic materials proposed by Lopez-Pamies

$$\psi^{\text{Eq}}(\mathbf{F}) = \begin{cases} \frac{3^{1-\alpha_1}}{2\alpha_1} \mu_1 [I_1^{\alpha_1} - 3^{\alpha_1}] + \frac{3^{1-\alpha_2}}{2\alpha_2} \mu_2 [I_1^{\alpha_2} - 3^{\alpha_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases}$$



$$\psi^{\text{NEq}}(\mathbf{F}^e) = \begin{cases} \frac{3^{1-a_1}}{2a_1} m_1 [I_1^{a_1} - 3^{a_1}] + \frac{3^{1-a_2}}{2a_2} m_2 [I_1^{a_2} - 3^{a_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases}$$

$$\phi(\mathbf{F}, \mathbf{F}^v, \dot{\mathbf{F}}^v) = \frac{1}{2} \tilde{\Gamma}^v \cdot \mathcal{A}(\mathbf{F}, \mathbf{F}^e) \tilde{\Gamma}^v$$

$$\tilde{\Gamma}^v = \mathbf{F}^e \Gamma^v \mathbf{F}^{e-1}, \quad \Gamma^v = \dot{\mathbf{F}}^v \mathbf{F}^{v-1}, \quad \mathcal{A}_{ijkl}(\mathbf{F}, \mathbf{F}^e) = F_{mi}^{e-1} F_{jn} A_{mnrs}(\mathbf{F}, \mathbf{F} \mathbf{F}^{v-1}) F_{rk}^{e-1} F_{ls}$$

$$\mathbf{S} = \frac{\partial \psi^{\text{Eq}}}{\partial \mathbf{F}}(\mathbf{F}) + \frac{\partial \psi^{\text{NEq}}}{\partial \mathbf{F}}(\mathbf{F} \mathbf{F}^{v-1})$$



$$\mathbf{S} = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r I_1^{\alpha_r-1} \right] \mathbf{F} + \left[\sum_{r=1}^2 3^{1-a_r} m_r \left(\mathbf{C} \cdot \mathbf{C}^{v-1} \right)^{a_r-1} \right] \mathbf{F} \mathbf{C}^{v-1} - p \mathbf{F}^{-T}$$

$$\frac{\partial \psi^{\text{NEq}}}{\partial \mathbf{F}^v}(\mathbf{F} \mathbf{F}^{v-1}) + \frac{\partial \phi}{\partial \dot{\mathbf{F}}^v}(\mathbf{F}, \mathbf{F}^v, \dot{\mathbf{F}}^v) = 0$$



$$\dot{\mathbf{C}}^v = \frac{\sum_{r=1}^2 3^{1-a_r} m_r \left(\mathbf{C} \cdot \mathbf{C}^{v-1} \right)^{a_r-1}}{3\eta_k} \left(\mathbf{C} - \frac{1}{3} \left(\mathbf{C} \cdot \mathbf{C}^{v-1} \right) \mathbf{C}^v \right)$$

The two-potential constitutive modeling of rubber viscoelastic materials proposed by Lopez-Pamies

$$\psi^{\text{Eq}}(\mathbf{F}) = \begin{cases} \frac{3^{1-\alpha_1}}{2\alpha_1} \mu_1 [I_1^{\alpha_1} - 3^{\alpha_1}] + \frac{3^{1-\alpha_2}}{2\alpha_2} \mu_2 [I_1^{\alpha_2} - 3^{\alpha_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases} \quad \longrightarrow \quad \mathbf{S} = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r I_1^{\alpha_r-1} \right] \mathbf{F} + \left[\sum_{r=1}^2 3^{1-a_r} m_r (\mathbf{C} \cdot \mathbf{C}^{\mathbf{v}-1})^{a_r-1} \right] \mathbf{F} \mathbf{C}^{\mathbf{v}-1} - p \mathbf{F}^{-\mathbf{T}}$$

$$\psi^{\text{NEq}}(\mathbf{F}^e) = \begin{cases} \frac{3^{1-a_1}}{2a_1} m_1 [I_1^{a_1} - 3^{a_1}] + \frac{3^{1-a_2}}{2a_2} m_2 [I_1^{a_2} - 3^{a_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases} \quad \dot{\mathbf{C}}^{\mathbf{v}} = \frac{\sum_{r=1}^2 3^{1-a_r} m_r (\mathbf{C} \cdot \mathbf{C}^{\mathbf{v}-1})^{a_r-1}}{3\eta_k} \left(\mathbf{C} - \frac{1}{3} (\mathbf{C} \cdot \mathbf{C}^{\mathbf{v}-1}) \mathbf{C}^{\mathbf{v}} \right)$$

$$S = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r \left(\frac{2}{\lambda} + \lambda^2 \right)^{\alpha_r-1} \right] \left(\lambda - \frac{1}{\lambda^2} \right) + \left[\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1} \right] \left(\frac{\lambda}{\lambda_v^2} - \frac{\lambda_v}{\lambda^2} \right) \quad \longleftarrow \text{Stress-deformation response for the uniaxial strain-controlled deformation}$$

$$\dot{\lambda}_v = \frac{\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1}}{3\eta_k} \left(\frac{\lambda^2}{\lambda_v} - \frac{\lambda_v^2}{\lambda} \right) \quad \longleftarrow \lambda_v \text{ is a solution of this ODE to be solved with Runge-Kutta method}$$

$$\eta_k = \eta_\infty + \frac{\eta_0 - \eta_\infty + k_1 \left[\left(\frac{2}{\lambda_v} + \lambda_v^2 \right)^{\beta_1} - 3^{\beta_1} \right]}{1 + \left(k_2 J_2^{\text{NEq}} \right)^{\beta_2}} \quad J_2^{\text{NEq}} = \frac{1}{3} \left(\frac{\lambda_v}{\lambda} - \frac{\lambda^2}{\lambda_v^2} \right)^2 \left(\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1} \right)^2$$

Runge-Kutta Method

$$\mathbf{C}_{t=t_{n+1}}^v = \mathbf{C}_{t=t_n}^v + \frac{\Delta t}{90} (7\mathbf{k}_1 + 32\mathbf{k}_3 + 12\mathbf{k}_4 + 32\mathbf{k}_5 + 7\mathbf{k}_6)$$

with

$$\mathbf{k}_1 = \mathbf{G}(t_n, \mathbf{C}_{t_n}^v)$$

$$\mathbf{k}_2 = \mathbf{G}(t_n + \Delta t/2, \mathbf{C}_{t_n}^v + \mathbf{k}_1 \Delta t/2)$$

$$\mathbf{k}_3 = \mathbf{G}(t_n + \Delta t/4, \mathbf{C}_{t_n}^v + (3\mathbf{k}_1 + \mathbf{k}_2) \Delta t/16)$$

$$\mathbf{k}_4 = \mathbf{G}(t_n + \Delta t/2, \mathbf{C}_{t_n}^v + \mathbf{k}_3 \Delta t/2)$$

$$\mathbf{k}_5 = \mathbf{G}(t_n + 3\Delta t/4, \mathbf{C}_{t_n}^v + 3(-\mathbf{k}_2 + 2\mathbf{k}_3 + 3\mathbf{k}_4) \Delta t/16)$$

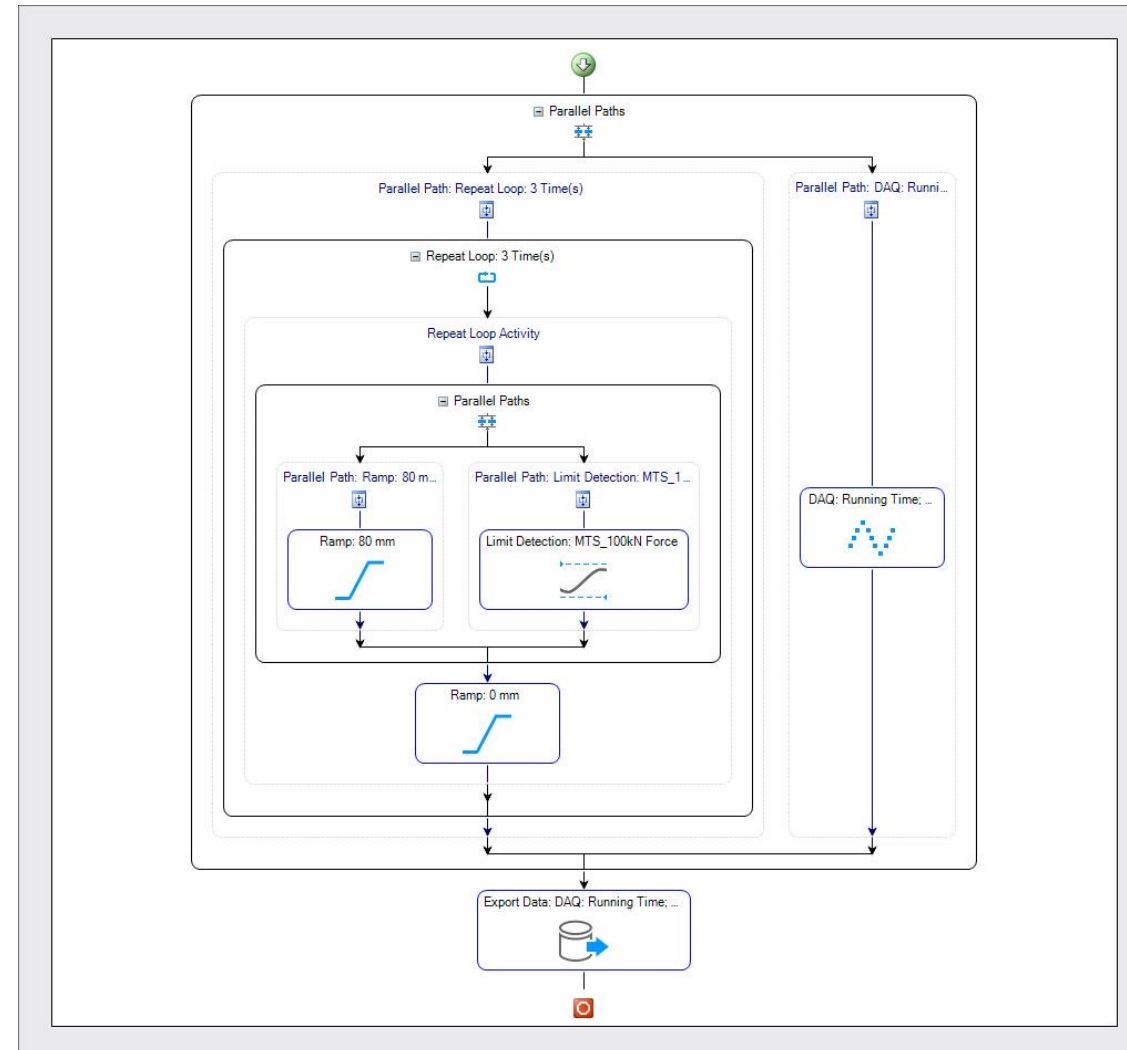
$$\mathbf{k}_6 = \mathbf{G}(t_n + \Delta t, \mathbf{C}_{t_n}^v + (\mathbf{k}_1 + 4\mathbf{k}_2 + 6\mathbf{k}_3 - 12\mathbf{k}_4 + 8\mathbf{k}_5) \Delta t/7)$$

Runge-Kutta Method implemented in MATLAB

```
function1[tt_] := (K1 = 0; K2 = 0; K3 = 0; K4 = 0; K5 = 0; K6 = 0;

λ3v = 1; dtime = 1;
tn = 0;
While[tn < tt,
  K1 = g[tn, λ3v];
  K2 = g[tn +  $\frac{dtime}{2}$ , λ3v +  $\frac{dtime}{2}$  K1];
  K3 = g[tn +  $\frac{dtime}{4}$ , λ3v +  $\frac{dtime}{16}$  (3 K1 + K2)];
  K4 = g[tn +  $\frac{dtime}{2}$ , λ3v +  $\frac{dtime}{16}$  (K3)];
  K5 = g[tn +  $\frac{3 dtime}{4}$ , λ3v +  $\frac{dtime}{16}$  3 (-K2 + 2 K3 + 3 K4)];
  K6 = g[tn + dtime, λ3v +  $\frac{dtime}{7}$  (K1 + 4 K2 + 6 K3 - 12 K4 + 8 K5)];
  λ3v = λ3v +  $\frac{dtime}{90}$  (7 K1 + 32 K3 + 12 K4 + 32 K5 + 7 K6); tn = tn + 1;]; S33num1 = S33[tt, λ3v])
```

Tensile MTS Program



Tensile MTS Program

